

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester III Examination, 2023-24

Paper No. MTB-302: Differential Equations

Time: 3 Hours

Full Marks: 70

Ref. No.: 067772

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Show that solutions of n^{th} -order differential equations may contain n arbitrary constants.
- (b) Solve the BVP: $\frac{d^2 y}{dx^2} + y = 0$, $y(0) = 1$, $y(\pi/2) = 0$.
- (c) Separate the variables in $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ and solve.
- (d) Solve by Bernoulli's equation: $\frac{dy}{dx} - \frac{y}{x} = y^2 \sin x$.
- (e) Define an indicial equation in ODE.
- (f) Show that $P_1(x) = x$ where $P_n(x)$ is the Legendre polynomial.
- (g) Show that $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$.

Question 2.

- (a) Reduce to variable-separable form and solve: $\cos(x+y) dx = dy$. [7]
- (b) Show that the necessary and sufficient condition for $M dx + N dy = 0$ to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. [7]

Question 3.

- (a) Solve (solvable for x): $y = 2px + y^2 p^2$ where $p = dy/dx$. [7]
- (b) Find the general and singular solution of $y = px + \sqrt{a^2 p^2 + b^2}$. [7]

Question 4.

(a) Solve: $(D^2 - 2D + 1)y = xe^x \sin x$. [7]

(b) Solve: $(D^2 + 1)y = \sin x - \sin 2x$. [7]

Question 5.

(a) Solve: $x^2 \frac{d^2 y}{dx^2} - 2y \frac{dy}{dx} = x^2$. [7]

(b) Change the independent variable in $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - 4y^2 \frac{d^2 y}{dx^2} + 4y \frac{dy}{dx} + \sin x^2$ and solve. [7]

Question 6. Solve in series:

$$(x - x^2)y'' + (1 - 5x)y' - 4y = 0$$

[14]

Question 7. Show that:

(a) $\int_{-1}^1 P_m(x)P_n(x) dx = 0$ if $m \neq n$. [7]

(b) $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$ and $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$. [7]